

# COS1501 - STUDY UNIT 10

Logic: Quantifiers, Predicates and Proof Strategies

Full source-aligned coverage

Medium + hard worked examples

Exam methods

Final revision

Based on the official UNISA COS1501 Study Unit 10 structure (pp. 151-165 of the study guide). Explanations, worked examples, memory rules and revision questions are newly written to make the unit easier to learn without replacing the terminology used in the source.

# Unit 10 Roadmap

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1. What Unit 10 is doing
2. Variables, open statements and truth values
3. Universal quantifier  $\forall$
4. Existential quantifier  $\exists$
5. Quantified variables as dummy variables
6. Quantifiers as generalised AND / OR
7. Predicates  $P(x)$
8. Negating quantified statements
9. Negating compound predicates and implications
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## Core idea of Unit 10

Unit 9 told you how to calculate the truth value of compound statements. Unit 10 moves from propositions such as  $p$  and  $q$  to statements containing variables, then shows how to quantify them and how to prove them when checking every possible value is impossible.

# 1. What Unit 10 is really about

The official unit asks you to understand universal and existential quantifiers, predicates, implication, counterexamples, converse, contrapositive, contradiction and vacuous proof. It then expects you to use those ideas to prove or disprove quantified mathematical statements.

## Prerequisite from Unit 9

Remember:  $p \rightarrow q$  is false only when  $p$  is true and  $q$  is false. Also  $p \rightarrow q \equiv \neg p \vee q$ . De Morgan's laws are  $\neg(p \wedge q) \equiv \neg p \vee \neg q$  and  $\neg(p \vee q) \equiv \neg p \wedge \neg q$ . These are used constantly in Unit 10.

# 2. Variables, open statements and truth values

A sentence such as “**x is even**” is not yet a complete declarative statement if  $x$  has not been given a value or quantified. Its truth depends on  $x$ . You can turn it into a statement with a truth value in two ways: substitute a particular value, or quantify the variable.

Open statement:  $P(x)$ : “ $x$  is even”

Substitution:  $P(8)$  is true;  $P(7)$  is false.

Quantification:  $\forall x \in \mathbb{Z}, P(x)$  or  $\exists x \in \mathbb{Z}, P(x)$ .

## MEDIUM - Turning an open statement into a statement

**Question.** Let  $P(x)$  mean “ $x^2 = 16$ ” with  $x \in \mathbb{Z}$ . Give two substitutions and their truth values.

**Solution.**  $P(4)$  is true because  $4^2 = 16$ .  $P(5)$  is false because  $5^2 = 25$ . The variable has been replaced by a specific integer, so each result now has a definite truth value.

## HARD - Domain changes the truth

**Question.** Let  $Q(x)$  mean “ $x^2 = 2$ ”. Compare  $\exists x \in \mathbb{Q}, Q(x)$  with  $\exists x \in \mathbb{R}, Q(x)$ .

**Solution.** Over  $\mathbb{Q}$  the statement is false because  $\sqrt{2}$  is not rational. Over  $\mathbb{R}$  it is true because  $x = \sqrt{2}$  is a real witness. The same predicate can change truth value when the domain changes.

## Exam trap

Never ignore the domain.  $\exists x \in \mathbb{Z}, x^2 = 2$  and  $\exists x \in \mathbb{R}, x^2 = 2$  are different statements.

# 3. Universal quantifier $\forall$ - “for all”

The universal quantifier  $\forall$  means that the stated property must hold for every element in the domain. Phrases such as “for all”, “for every” and “for each” are universal quantification.

$\forall x \in A, P(x)$  means: every  $x$  in  $A$  has property  $P$ .

### Truth rule

A universal statement is true only if there is **no exception**. One exception is enough to make it false.

### MEDIUM - Universal statement on a finite set

**Question.** Let  $A = \{2, 4, 6, 8\}$ . Decide whether  $\forall x \in A, x$  is even.

**Solution.** Check every element: 2, 4, 6 and 8 are all even. Therefore the universal statement is true.

### HARD - Universal statement over infinitely many integers

**Question.** Determine whether  $\forall n \in \mathbb{Z}, n^2 + n$  is even.

**Solution.** Write  $n^2 + n = n(n+1)$ . Two consecutive integers always contain one even integer, so their product is even. Therefore the statement is true for every integer  $n$ .

## 4. Existential quantifier $\exists$ - “there exists”

The existential quantifier  $\exists$  means that at least one element in the domain has the required property. Phrases such as “there exists”, “for some” and “we can find” are existential quantification.

$\exists x \in A, P(x)$  means: at least one  $x$  in  $A$  has property  $P$ .

### Truth rule

To prove an existential statement, one valid example - called a **witness** - is enough.

### MEDIUM - Finding a witness

**Question.** Determine whether  $\exists x \in \mathbb{Z}, x^2 = 49$ .

**Solution.** Yes.  $x = 7$  is a witness because  $7^2 = 49$ .  $x = -7$  is another witness, but one is enough.

### HARD - Existence with two conditions

**Question.** Determine whether  $\exists n \in \mathbb{Z}^+$  such that  $n$  is a multiple of 6 and  $n^2 < 100$ .

**Solution.** Choose  $n = 6$ . It is a positive multiple of 6 and  $6^2 = 36 < 100$ . Therefore the existential statement is true.

## 5. Quantified variables are dummy variables

A quantified variable is only a placeholder. Renaming it everywhere does not change the meaning, provided the new name does not cause a clash with another variable.

$$\forall x \in \mathbb{R}, x^2 \geq 0 \equiv \forall t \in \mathbb{R}, t^2 \geq 0$$

### MEDIUM - Renaming safely

**Question.** Rewrite  $\exists x \in \mathbb{Z}, x + 5 = 0$  using  $y$  instead of  $x$ .

**Solution.**  $\exists y \in \mathbb{Z}, y + 5 = 0$ . Nothing mathematical changed; only the placeholder name changed.

### HARD - Renaming in a two-variable statement

**Question.** Rewrite  $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, y > x$  using  $m$  and  $n$ .

**Solution.**  $\forall m \in \mathbb{Z}, \exists n \in \mathbb{Z}, n > m$ . Rename consistently:  $x \rightarrow m$  and  $y \rightarrow n$ . The order of the quantifiers stays the same.

#### Important

Renaming a variable is harmless. Swapping the order of quantifiers can completely change meaning. For example,  $\forall x \exists y (y > x)$  is true over  $\mathbb{Z}$ , but  $\exists y \forall x (y > x)$  is false.

## 6. Quantifiers as generalised conjunction and disjunction

### 6.1 Universal quantification behaves like a long AND

For a finite set  $A = \{a_1, a_2, \dots, a_n\}$ , the statement  $\forall x \in A, P(x)$  has the same truth behaviour as  $P(a_1) \wedge P(a_2) \wedge \dots \wedge P(a_n)$ . Every component must be true.

#### MEDIUM - Generalised AND

**Question.** Let  $A = \{1, 2, 3\}$ . Evaluate  $\forall x \in A, x < 4$ .

**Solution.** This is  $(1 < 4) \wedge (2 < 4) \wedge (3 < 4)$ . All three are true, so the quantified statement is true.

#### HARD - Finding the failing component

**Question.** Let  $A = \{-2, -1, 0, 1, 2\}$ . Evaluate  $\forall x \in A, x^2 > 0$ .

**Solution.** The expanded conjunction contains  $0^2 > 0$ , which is false. Therefore the entire universal statement is false.  $x = 0$  is the counterexample.

### 6.2 Existential quantification behaves like a long OR

For a finite set,  $\exists x \in A, P(x)$  behaves like  $P(a_1) \vee P(a_2) \vee \dots \vee P(a_n)$ . At least one component must be true.

### MEDIUM - Generalised OR

**Question.** Let  $A=\{1,2,3\}$ . Evaluate  $\exists x \in A, x^2=4$ .

**Solution.** This is  $(1^2=4) \vee (2^2=4) \vee (3^2=4)$ . The middle component is true, so the existential statement is true.

### HARD - Existential statement with no witness

**Question.** Let  $A=\{-3,-2,-1,0,1,2,3\}$ . Evaluate  $\exists x \in A, x^2=10$ .

**Solution.** No listed element squares to 10. Every component of the generalised OR is false, so the existential statement is false.

### Why quantifiers matter

For infinite domains you cannot literally write an infinite AND or OR. Quantifiers let you express the idea compactly and proofs let you establish truth without checking infinitely many cases.

## 7. Predicates $P(x)$

A predicate is a statement form  $P(x)$  expressing a property of a variable  $x$  in a domain. After  $x$  is given a value,  $P(x)$  is either true or false. In computer-science language, you can think of a predicate as a Boolean test.

$P(n)$ : “ $n$  is divisible by 3”

$P(12)$  = True;  $P(14)$  = False

### MEDIUM - Evaluate a predicate

**Question.** Let  $P(x): x^2 - 1 = 0$  over  $\mathbb{R}$ . Evaluate  $P(1)$ ,  $P(-1)$ ,  $P(2)$ .

**Solution.**  $P(1)$  is true;  $P(-1)$  is true;  $P(2)$  is false.

### HARD - Build a predicate from an English condition

**Question.** Define a predicate for “ $n$  is an odd multiple of 5”, with  $n \in \mathbb{Z}$ , and test  $n=15$  and  $n=20$ .

**Solution.** One form is  $P(n): (5 \text{ divides } n) \wedge (n \text{ is odd})$ .  $P(15)$  is true because 15 is divisible by 5 and odd.  $P(20)$  is false because it is divisible by 5 but even.

## 8. Negation of quantified statements

Negating a quantifier switches the quantifier and negates the predicate. This is one of the most important rules in the unit.

$\neg(\forall x \in A, P(x)) \equiv \exists x \in A, \neg P(x)$

$\neg(\exists x \in A, P(x)) \equiv \forall x \in A, \neg P(x)$

### Memory rule

**NOT ALL = SOME NOT.**

**NOT SOME = ALL NOT.**

When  $\neg$  crosses a quantifier:  $\forall$  flips to  $\exists$ , and  $\exists$  flips to  $\forall$ .

### MEDIUM - Negating a universal inequality

**Question.** Negate:  $\forall x \in \mathbb{R}, x > 5$ .

**Solution.**  $\exists x \in \mathbb{R}$  such that  $x \leq 5$ . The quantifier flips  $\forall \rightarrow \exists$  and " $x > 5$ " becomes " $x \leq 5$ ".

### HARD - Negating an existential equation

**Question.** Negate:  $\exists n \in \mathbb{Z}, n^2 + n = 7$ .

**Solution.**  $\forall n \in \mathbb{Z}, n^2 + n \neq 7$ . The quantifier flips  $\exists \rightarrow \forall$  and equality is negated to inequality.

## 9. Negating inequalities, compound predicates and implications

### 9.1 Negating comparison symbols

| Statement  | Negation   |
|------------|------------|
| $x > a$    | $x \leq a$ |
| $x \geq a$ | $x < a$    |
| $x < a$    | $x \geq a$ |
| $x \leq a$ | $x > a$    |
| $x = a$    | $x \neq a$ |
| $x \neq a$ | $x = a$    |

### MEDIUM - Inequality negation

**Question.** Negate  $x^3 \geq 8$ .

**Solution.**  $x^3 < 8$ .

### HARD - Compound comparison

**Question.** Negate  $(x > 0) \wedge (x \leq 4)$ .

**Solution.** By De Morgan:  $(x \leq 0) \vee (x > 4)$ .

## 9.2 De Morgan inside quantifiers

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q$$

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q$$

### MEDIUM - Quantifier + AND

**Question.** Negate:  $\exists x \in \mathbb{Z}, (x > 2) \wedge (x \text{ is even})$ .

**Solution.**  $\forall x \in \mathbb{Z}, (x \leq 2) \vee (x \text{ is not even})$ .

### HARD - Quantifier + OR

**Question.** Negate:  $\forall x \in \mathbb{R}, (x < 0) \vee (x^2 \geq 1)$ .

**Solution.**  $\exists x \in \mathbb{R}, (x \geq 0) \wedge (x^2 < 1)$ . Every layer of the negation has been pushed inward.

## 9.3 Negating an implication under a quantifier

A very useful identity is  $\neg(P \rightarrow Q) \equiv P \wedge \neg Q$ . The only way an implication fails is when its antecedent is true and its consequent is false.

$$\neg[\forall x \in A, P(x) \rightarrow Q(x)]$$

$$\equiv \exists x \in A, \neg(P(x) \rightarrow Q(x))$$

$$\equiv \exists x \in A, P(x) \wedge \neg Q(x)$$

### MEDIUM - Negate a quantified implication

**Question.** Negate:  $\forall n \in \mathbb{Z}, (n \text{ is divisible by } 4) \rightarrow (n \text{ is even})$ .

**Solution.**  $\exists n \in \mathbb{Z}$  such that  $n$  is divisible by 4 and  $n$  is not even.

### HARD - Negate with inequalities

**Question.** Negate:  $\forall x \in \mathbb{R}, (x \geq 2) \rightarrow (x^2 \geq 4)$ .

**Solution.**  $\exists x \in \mathbb{R}$  such that  $(x \geq 2) \wedge (x^2 < 4)$ .

## 10. Witnesses and counterexamples

A **witness** proves an existential statement. A **counterexample** disproves a universal statement. These are two sides of the same logic because the negation of a universal statement is existential.

To prove  $\exists x \in A, P(x)$ : find one witness  $a \in A$  with  $P(a)$  true.

To disprove  $\forall x \in A, P(x)$ : find one counterexample  $a \in A$  with  $P(a)$  false.



### MEDIUM - Counterexample

**Question.** Disprove  $\forall n \in \mathbb{Z}, n^2 > n$ .

**Solution.** Take  $n=0$ . Then  $0^2 > 0$  is false. One counterexample disproves the universal claim.

### HARD - Counterexample to an implication

**Question.** Disprove  $\forall x \in \mathbb{R}, (x > 0) \rightarrow (x^2 - 5x + 4 < 0)$ .

**Solution.** Choose  $x=5$ . The antecedent  $5 > 0$  is true, but  $25 - 25 + 4 = 4 < 0$  is false. Therefore the implication fails at  $x=5$ .

## 11. Proof strategies - the big picture

A proof strategy is a standard logical route for establishing a statement without checking all possible cases individually. The official unit focuses on four strategies: direct proof, contradiction, contrapositive and vacuous proof. It also shows how these methods apply to quantified statements.

| Goal                             | Best starting idea                                 |
|----------------------------------|----------------------------------------------------|
| Prove $p \rightarrow q$ directly | Assume $p$ ; derive $q$ .                          |
| Proof by contradiction           | Assume $p$ and $\neg q$ ; derive an impossibility. |
| Proof by contrapositive          | Prove $\neg q \rightarrow \neg p$ instead.         |
| Disprove a universal claim       | Find one counterexample.                           |
| Prove an existential claim       | Give one witness.                                  |
| Vacuous proof                    | Show the antecedent can never be true.             |

### Important

Different methods can prove the same theorem. In an exam, choose the method that makes the algebra or logical structure shortest and clearest.

## 12. Direct proof

To prove  $p \rightarrow q$  directly: assume  $p$  is true, use definitions and known facts, and reach  $q$ . Do not assume  $q$ .

Direct proof template:

1. Let the variables be arbitrary elements of the stated domain.
2. Assume  $p$ .
3. Rewrite  $p$  using definitions.
4. Perform valid algebra/logical steps.
5. Reach  $q$ .
6. Therefore  $p \rightarrow q$ .

### MEDIUM - Parity

**Question.** Prove: for every  $n \in \mathbb{Z}$ , if  $n$  is even, then  $n^2 + 6n$  is even.

**Solution.** Assume  $n$  is even. Then  $n = 2k$  for some  $k \in \mathbb{Z}$ . So  $n^2 + 6n = (2k)^2 + 6(2k) = 4k^2 + 12k = 2(2k^2 + 6k)$ . The bracket is an integer, so the expression is even.

### **HARD - Factorised inequality**

**Question.** Prove: if  $x^2 - 7x + 10 < 0$ , then  $x > 2$ .

**Solution.** Factor:  $(x-5)(x-2) < 0$ . A product is negative when the factors have opposite signs. The possible consistent case is  $x-5 < 0$  and  $x-2 > 0$ , giving  $2 < x < 5$ . Hence  $x > 2$ .

## **13. Proof by contradiction - reductio ad absurdum**

To prove  $p \rightarrow q$  by contradiction, assume  $p$  is true and also assume the bad possibility  $\neg q$ . If these assumptions force an impossibility,  $\neg q$  cannot be true, so  $q$  must be true.

Contradiction template:

Assume  $p$ .

Assume  $\neg q$ .

Derive  $r$  and  $\neg r$ , or some impossible fact.

Therefore  $\neg q$  is impossible; hence  $q$ .

### **MEDIUM - Parity contradiction**

**Question.** Prove: if  $n^2$  is odd, then  $n$  is odd.

**Solution.** Assume  $n^2$  is odd. Suppose for contradiction that  $n$  is not odd, so  $n$  is even. Then  $n = 2k$ , so  $n^2 = 4k^2 = 2(2k^2)$ , which is even. That contradicts the assumption that  $n^2$  is odd. Hence  $n$  is odd.

### **HARD - No greatest integer**

**Question.** Prove that there is no greatest integer.

**Solution.** Suppose for contradiction that  $g$  is the greatest integer. Then  $g+1$  is also an integer, and  $g+1 > g$ . This contradicts the claim that  $g$  is greatest. Therefore no greatest integer exists.

### **What counts as a contradiction?**

Examples: obtaining  $0=1$ ; showing a number is both even and odd; deriving  $x > 5$  and  $x \leq 5$  simultaneously; contradicting a stated assumption.

## **14. Proof by contrapositive**

The contrapositive of  $p \rightarrow q$  is  $\neg q \rightarrow \neg p$ . These are logically equivalent, so proving the contrapositive proves the original implication.

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

### MEDIUM - Divisibility

**Question.** Prove: if  $n^2$  is even, then  $n$  is even.

**Solution.** Use contrapositive. Assume  $n$  is not even, so  $n$  is odd:  $n=2k+1$ . Then  $n^2=(2k+1)^2=4k^2+4k+1=2(2k^2+2k)+1$ , which is odd. Thus if  $n^2$  is not odd (i.e. is even),  $n$  cannot be odd;  $n$  is even.

### HARD - Inequality

**Question.** Prove: for real  $x$ , if  $x^2 < 9$  then  $-3 < x < 3$ .

**Solution.** Contrapositive: if  $x \leq -3$  or  $x \geq 3$ , then  $x^2 \geq 9$ . If  $x \geq 3$ , squaring gives  $x^2 \geq 9$ . If  $x \leq -3$ , then  $|x| \geq 3$ , so  $x^2 = |x|^2 \geq 9$ . Therefore the original implication holds.

## 15. Converse vs contrapositive

For an implication  $p \rightarrow q$ :

| Name           | Form                        | Equivalent to original? |
|----------------|-----------------------------|-------------------------|
| Original       | $p \rightarrow q$           | Yes - it is itself      |
| Converse       | $q \rightarrow p$           | Not in general          |
| Contrapositive | $\neg q \rightarrow \neg p$ | Yes                     |
| Inverse        | $\neg p \rightarrow \neg q$ | Not in general          |

### MEDIUM - A false converse

**Question.** Original: If  $n$  is divisible by 4, then  $n$  is even. Write the converse and test it.

**Solution.** Converse: If  $n$  is even, then  $n$  is divisible by 4. False:  $n=6$  is even but not divisible by 4. The original can be true while the converse is false.

### HARD - Identify all forms

**Question.** Let  $p$ : " $x > 3$ " and  $q$ : " $x^2 > 9$ ". Write converse and contrapositive.

**Solution.** Original:  $x > 3 \rightarrow x^2 > 9$ . Converse:  $x^2 > 9 \rightarrow x > 3$  (false for  $x = -4$ ). Contrapositive:  $x^2 \leq 9 \rightarrow x \leq 3$  (true and logically equivalent to the original).

### Exam trap

Do not prove the converse when asked to prove the original implication. The contrapositive is safe; the converse is a different statement.

## 16. Proofs involving universal quantifiers

To prove  $\forall x \in A, P(x)$ , choose an arbitrary  $x$  in  $A$  and prove  $P(x)$  without using any special property that only one particular element has. For a statement written with membership as an implication, think “if  $x \in A$ , then  $P(x)$ ”.

Universal proof template:  
Let  $x$  be arbitrary in  $A$ .  
Use only facts true for every  $x \in A$ .  
Prove  $P(x)$ .  
Therefore  $\forall x \in A, P(x)$ .

### MEDIUM - Positive expression

**Question.** Prove  $\forall x \in \mathbb{R}, x^2 + 4 > 0$ .

**Solution.** Let  $x$  be any real number. Since  $x^2 \geq 0$ ,  $x^2 + 4 \geq 4 > 0$ . Therefore the statement holds for every real  $x$ .

### HARD - Divisibility identity

**Question.** Prove  $\forall n \in \mathbb{Z}, n^3 - n$  is divisible by 3.

**Solution.** Factor  $n^3 - n = n(n-1)(n+1)$ , the product of three consecutive integers. Among any three consecutive integers, one is divisible by 3. Hence the product is divisible by 3 for every integer  $n$ .

## 17. Proving existential statements

For  $\exists x \in A, P(x)$ , you normally do not need a general proof. Supply a valid element and verify the property. The chosen element is the witness.

### MEDIUM - Simple witness

**Question.** Prove  $\exists n \in \mathbb{Z}, n^2 + n = 6$ .

**Solution.** Choose  $n=2$ . Then  $2^2 + 2 = 6$ . Therefore such an integer exists.

### HARD - Constructing a witness

**Question.** Prove: for every integer  $a$ , there exists an integer  $b$  such that  $b > a$ .

**Solution.** Let  $a$  be arbitrary. Choose  $b = a + 1$ . Since  $a \in \mathbb{Z}$ ,  $b \in \mathbb{Z}$ , and  $a + 1 > a$ . Therefore  $\forall a \in \mathbb{Z}, \exists b \in \mathbb{Z}, b > a$ .

### Quantifier order matters

The hard example has  $\forall a \exists b$ . Your witness  $b$  is allowed to depend on  $a$ . This is different from  $\exists b \forall a$ , which would claim one single integer exceeds every integer.

## 18. Disproving a universal quantified statement

If a universal statement is false, do not try to list many failures. One counterexample is logically enough.

To disprove  $\forall x \in A, P(x)$ , prove  $\exists x \in A, \neg P(x)$ .

### MEDIUM - Polynomial counterexample

**Question.** Disprove  $\forall x \in \mathbb{R}, x^2 - 4x + 3 \geq 0$ .

**Solution.** Take  $x=2$ . Then  $4-8+3=-1 < 0$ . So the universal statement is false.

### HARD - Implication counterexample

**Question.** Disprove  $\forall n \in \mathbb{Z}, (n^2 \text{ is divisible by } 4) \rightarrow (n \text{ is divisible by } 4)$ .

**Solution.** Choose  $n=2$ . Then  $n^2=4$  is divisible by 4, but  $n=2$  is not divisible by 4. Antecedent true + consequent false gives a counterexample.

## 19. Vacuous proof

An implication  $p \rightarrow q$  is automatically true whenever  $p$  is false. A vacuous proof shows that the antecedent can never happen in the relevant situation, so there is no possible counterexample to the implication.

If  $p$  is impossible/always false, then  $p \rightarrow q$  is true regardless of  $q$ .

### MEDIUM - Empty set subset

**Question.** Prove  $\emptyset \subseteq X$  for every set  $X$ .

**Solution.** By definition, we need: if  $x \in \emptyset$ , then  $x \in X$ . But  $x \in \emptyset$  is impossible because  $\emptyset$  has no elements. The antecedent is always false, so the implication is vacuously true. Hence  $\emptyset \subseteq X$ .

### HARD - Transitivity with no chain

**Question.** Let  $S = \{(a,b), (a,d)\}$  on  $\{a,b,c,d\}$ . Show  $S$  is transitive.

**Solution.** Transitivity only requires action when  $(x,y)$  and  $(y,z)$  are both in  $S$ . No pair starts with  $b$  or  $d$ , so no two pairs can form such a chain. The antecedent of the transitivity condition never occurs; therefore  $S$  is vacuously transitive.

### Vacuous does not mean “fake”

Vacuous truth is ordinary implication logic. It says there is no case where the hypothesis is true and the conclusion false.

## 20. How to choose the best proof method

| Clue in the question                                   | Usually try                                 |
|--------------------------------------------------------|---------------------------------------------|
| "If $n$ is even / divisible by $k$ ..."                | Direct proof: substitute $n=2m$ or $n=km$ . |
| Conclusion is a negative statement such as $x \neq 1$  | Contrapositive may be extremely short.      |
| Assuming the opposite creates an obvious impossibility | Contradiction.                              |
| Original algebra is awkward but negations are simple   | Contrapositive.                             |
| Universal claim looks false                            | Counterexample.                             |
| Existential claim                                      | Find/construct a witness.                   |
| Antecedent can never occur                             | Vacuous proof.                              |

### Proof-writing standard

State the domain. State your assumption. Name the definition you use (even =  $2k$ , multiple of 3 =  $3k$ , etc.). Make every algebraic step valid. End by explicitly connecting the result to what had to be shown.

## 21. Official Study Unit 10 activity walkthrough

The following section follows the skill types in Activities 10-3 to 10-10 of the official unit. The wording is shortened, but every tested idea is covered.

### 21.1 Activity 10-3: translate and judge quantified statements

| Statement                                | English meaning                                  | Truth                                        |
|------------------------------------------|--------------------------------------------------|----------------------------------------------|
| $\exists y \in \mathbb{Q}, y = \sqrt{2}$ | There is a rational number equal to $\sqrt{2}$ . | False: $\sqrt{2}$ is irrational.             |
| $\forall x \in \mathbb{R}, 2x < x^2$     | For every real $x$ , $2x$ is less than $x^2$ .   | False: $x=0$ gives $0 < 0$ , false.          |
| $\forall x \in \mathbb{Z}, x > 0$        | Every integer is positive.                       | False: $x=0$ (or $-1$ ) is a counterexample. |
| $\exists x \in \mathbb{Z}^+, x = 0$      | Some positive integer equals 0.                  | False: 0 is not positive.                    |

### 21.2 Activities 10-4 and 10-5: De Morgan truth-table facts

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

These equivalences justify pushing a negation through AND/OR when you negate quantified compound predicates.

### 21.3 Activity 10-6: negations - worked answers

| Original                                                          | Negation                                                   |
|-------------------------------------------------------------------|------------------------------------------------------------|
| $\forall x \in \mathbb{Z}^+, x > 3$                               | $\exists x \in \mathbb{Z}^+, x \leq 3$                     |
| $\exists x \in \mathbb{R}, 2x = x^2$                              | $\forall x \in \mathbb{R}, 2x \neq x^2$                    |
| $\forall x \in \mathbb{Z}, (x > 0) \vee (x^2 > 0)$                | $\exists x \in \mathbb{Z}, (x \leq 0) \wedge (x^2 \leq 0)$ |
| $\exists y \in \mathbb{Z}^+, (y \leq 10) \wedge (y \neq 0)$       | $\forall y \in \mathbb{Z}^+, (y > 10) \vee (y = 0)$        |
| $\exists x \in A, P(x) \wedge Q(x)$                               | $\forall x \in A, \neg P(x) \vee \neg Q(x)$                |
| $\forall x \in \mathbb{Z}^+, (x \leq 3) \rightarrow (x^3 \geq 1)$ | $\exists x \in \mathbb{Z}^+, (x \leq 3) \wedge (x^3 < 1)$  |

### 21.4 Activity 10-7: original or negation?

| Item                                                        | Result                         | Reason                        |
|-------------------------------------------------------------|--------------------------------|-------------------------------|
| $\forall x \in \mathbb{Z}^+, x > 3$                         | Original false; negation true. | $x=1$ is a counterexample.    |
| $\exists x \in \mathbb{R}, 2x = x^2$                        | Original true; negation false. | $x=0$ or $x=2$ is a witness.  |
| $\forall x \in \mathbb{Z}, (x > 0) \vee (x^2 > 0)$          | Original false; negation true. | $x=0$ makes both parts false. |
| $\exists y \in \mathbb{Z}^+, (y \leq 10) \wedge (y \neq 0)$ | Original true; negation false. | $y=1$ is a witness.           |



## 21.5 Activity 10-8: implication versus converse

| p | q | $p \rightarrow q$ | $q \rightarrow p$ |
|---|---|-------------------|-------------------|
| T | T | T                 | T                 |
| T | F | F                 | T                 |
| F | T | T                 | F                 |
| F | F | T                 | T                 |

The final two columns differ, so  $p \rightarrow q$  and  $q \rightarrow p$  are not logically equivalent.

## 21.6 Activity 10-9: counterexample

For the claim  $\forall x \in \mathbb{R}, x^2 - 4x + 3 \geq 0$ , choose  $x = 2$ . The expression is  $-1$ , so the universal claim is false. The official study guide uses another valid value; any one valid counterexample is sufficient.

## 21.7 Activity 10-10: proof-method practice

### (a) If $x^2 - 3x + 2 < 0$ , then $x > 0$

**Best method: direct.** Factor  $(x-1)(x-2) < 0$ , so  $1 < x < 2$ , hence  $x > 0$ . Contrapositive and contradiction also work, but are longer.

### (b) If $x^2 - x - 6 > 0$ , then $x \neq 1$

**Best method: contrapositive.** The contrapositive is: if  $x = 1$ , then  $x^2 - x - 6 \leq 0$ . Substituting  $x = 1$  gives  $-6 \leq 0$ , immediately true. Direct proof can also factor  $(x-3)(x+2) > 0$  to get  $x > 3$  or  $x < -2$ .

### (c) If $a + b$ is odd, exactly one of $a, b$ is odd

**Best method: contrapositive/cases.** If it is not true that exactly one is odd, then  $a$  and  $b$  have the same parity. Even+even is even and odd+odd is even, so  $a + b$  is not odd. Therefore the original implication holds.

### (d) If $x$ is even, then $x^2 + 4x + 2$ is even

**Best method: direct.** Let  $x = 2k$ . Then  $x^2 + 4x + 2 = 4k^2 + 8k + 2 = 2(2k^2 + 4k + 1)$ , which is even.

### (e) If $n$ is a multiple of 3, then $n^3 + n^2$ is a multiple of 3

**Best method: direct.** Let  $n = 3k$ . Then  $n^3 + n^2 = 27k^3 + 9k^2 = 3(9k^3 + 3k^2)$ , so it is divisible by 3.

### Counterexample item

For the claim “if  $x > 0$  then  $x^2 - 3x + 1 < 0$ ” for all positive integers  $x$ , take  $x = 3$ . The antecedent is true, but  $9 - 9 + 1 = 1 < 0$  is false. Therefore the universal implication is false.

## 22. More worked examples for deeper understanding

### EXTRA - Quantifier order

**Question.** Compare  $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, y = x + 1$  with  $\exists y \in \mathbb{Z}, \forall x \in \mathbb{Z}, y = x + 1$ .

**Solution.** The first is true: given any  $x$ , choose  $y = x + 1$ . The second is false: one fixed  $y$  cannot equal  $x + 1$  for every different integer  $x$ . Quantifier order changes meaning.

### EXTRA - Nested negation

**Question.** Negate  $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, y > x$ .

**Solution.** Flip each quantifier as the negation moves inward:  $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, \neg(y > x)$ , i.e.  $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}, y \leq x$ . This negation claims there is a greatest real number, which is false.

### EXTRA - Direct set proof phrased as implication

**Question.** Prove  $A \cap B \subseteq A$ .

**Solution.** Let  $x \in A \cap B$ . By definition of intersection,  $x \in A$  and  $x \in B$ . Therefore  $x \in A$ . Since  $x$  was arbitrary,  $A \cap B \subseteq A$ .

### EXTRA - Contrapositive in set language

**Question.** If  $A \subseteq B$  and  $x \notin B$ , prove  $x \notin A$ .

**Solution.** This is exactly the contrapositive of “if  $x \in A$  then  $x \in B$ ”, which is the elementwise meaning of  $A \subseteq B$ .

### EXTRA - Vacuous relation property

**Question.** Let  $R = \emptyset$  on a nonempty set  $A$ . Is  $R$  transitive?

**Solution.** Yes. Transitivity requires: if  $(x, y) \in R$  and  $(y, z) \in R$ , then  $(x, z) \in R$ . The antecedent can never be true because  $R$  has no ordered pairs, so the condition is vacuously true.

### EXTRA - Hard contradiction

**Question.** Prove  $\sqrt{3}$  is irrational.

**Solution.** Assume  $\sqrt{3} = p/q$  in lowest terms. Then  $p^2 = 3q^2$ , so 3 divides  $p^2$  and hence  $p$ ; write  $p = 3k$ . Then  $9k^2 = 3q^2$ , so  $q^2 = 3k^2$  and 3 divides  $q$ . Thus  $p$  and  $q$  share factor 3, contradicting lowest terms. Therefore  $\sqrt{3}$  is irrational.

## 23. Common mistakes that lose marks

- **Forgetting the domain:** truth can change between  $\mathbb{Z}$ ,  $\mathbb{Q}$  and  $\mathbb{R}$ .
- **Negating without flipping the quantifier:**  $\neg \forall$  is  $\exists \neg$ , not  $\forall \neg$ .

- **Forgetting De Morgan:** negating AND turns it into OR; negating OR turns it into AND.
- **Negating an implication incorrectly:**  $\neg(P \rightarrow Q)$  is  $P \wedge \neg Q$ .
- **Using the converse as proof:**  $q \rightarrow p$  is generally not equivalent to  $p \rightarrow q$ .
- **Giving many examples to prove a universal statement:** examples can suggest a theorem but do not prove infinitely many cases.
- **Giving a counterexample that does not satisfy the hypothesis:** to refute  $p \rightarrow q$  you need  $p$  true and  $q$  false.
- **Writing  $n=2k$  without saying  $k \in \mathbb{Z}$ :** the integer condition matters in parity proofs.
- **Assuming what you are trying to prove:** in a direct proof, assume only the antecedent.
- **Missing vacuous truth:** if no element satisfies the hypothesis, the implication is still true.

## 24. Final revision sheet

| Concept          | Must know instantly                            |
|------------------|------------------------------------------------|
| $\forall$        | for all / every / each                         |
| $\exists$        | there exists / some                            |
| Predicate        | $P(x)$ has a truth value after $x$ is fixed    |
| $\neg \forall P$ | $\exists \neg P$                               |
| $\neg \exists P$ | $\forall \neg P$                               |
| Counterexample   | one value disproves a universal claim          |
| Witness          | one value proves an existential claim          |
| Direct           | $p \Rightarrow \dots \Rightarrow q$            |
| Contradiction    | assume $p$ and $\neg q$ ; derive impossibility |
| Contrapositive   | prove $\neg q \rightarrow \neg p$              |
| Converse         | $q \rightarrow p$ ; not generally equivalent   |
| Vacuous          | antecedent impossible, so implication true     |

## 25. Practice test

1. Translate into symbols: "Every integer has a square that is non-negative."
2. Translate into English:  $\exists x \in \mathbb{Z}, x^3 = 27$ .
3. Negate:  $\forall x \in \mathbb{R}, x^2 + 1 > 0$ .
4. Negate:  $\exists n \in \mathbb{Z}, (n > 5) \wedge (n \text{ is even})$ .
5. Negate:  $\forall n \in \mathbb{Z}, (4|n) \rightarrow (2|n)$ .
6. Give a counterexample to  $\forall x \in \mathbb{R}, x^2 \geq x$ .
7. Prove directly: if  $n$  is odd, then  $n^2$  is odd.

8. Prove by contrapositive: if  $n^2$  is odd, then  $n$  is odd.
9. Write the converse and contrapositive of: if  $x > 5$  then  $x > 2$ .
10. Prove  $\forall x \in \mathbb{R}, x^2 - 2x + 2 > 0$ .
11. Prove  $\exists n \in \mathbb{Z}$  such that  $n^2 = 81$ .
12. Explain why  $\emptyset \subseteq A$  is vacuously true.
13. Determine whether  $\forall n \in \mathbb{Z}, n(n+1)$  is even.
14. Disprove: if  $n$  is prime then  $n$  is odd.
15. Explain the difference between  $\forall x \exists y (y > x)$  and  $\exists y \forall x (y > x)$  over  $\mathbb{Z}$ .

## 26. Practice test answers

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1.  $\forall n \in \mathbb{Z}, n^2 \geq 0$ .
2. There exists an integer whose cube is 27. True;  $x = 3$ .
3.  $\exists x \in \mathbb{R}, x^2 + 1 \leq 0$ .
4.  $\forall n \in \mathbb{Z}, (n \leq 5) \vee (n \text{ is odd})$ .
5.  $\exists n \in \mathbb{Z}, (4|n) \wedge \neg(2|n)$ .
6.  $x = 1/2$  gives  $1/4 \geq 1/2$ , false.
7. Let  $n = 2k + 1$ . Then  $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ , odd.
8. Contrapositive: if  $n$  is even then  $n^2$  is even. Let  $n = 2k$ ;  $n^2 = 4k^2 = 2(2k^2)$ .
9. Converse:  $x > 2 \rightarrow x > 5$  (false). Contrapositive:  $x \leq 2 \rightarrow x \leq 5$  (true).
10.  $x^2 - 2x + 2 = (x - 1)^2 + 1 > 0$  for every real  $x$ .
11. Choose  $n = 9$  (or  $-9$ ).
12. The condition  $x \in \emptyset$  is never true, so the implication  $x \in \emptyset \rightarrow x \in A$  cannot fail.
13. True: one of consecutive integers  $n, n+1$  is even.
14.  $n = 2$  is prime but not odd.
15.  $\forall x \exists y$  is true: choose  $y = x + 1$ .  $\exists y \forall x$  is false: there is no greatest integer.

## 27. Completion checklist

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- I can translate between English and  $\forall / \exists$  notation.
- I know that quantified variables are placeholders.
- I understand  $\forall$  as generalised AND and  $\exists$  as generalised OR.
- I can evaluate predicates and identify their domains.
- I can push negations through quantifiers, AND, OR and implications.
- I can prove an existential claim with a witness.
- I can disprove a universal claim with one counterexample.
- I can write direct, contradiction and contrapositive proofs.
- I do not confuse converse with contrapositive.
- I can recognise a vacuous proof.
- I can handle the exercise types in Activities 10-3 to 10-10.

### Bottom line

If you can do the checklist without looking at the notes and can solve the practice test cleanly, you have covered the full conceptual scope of Study Unit 10 in the source, plus extra exam-level practice.

## 28. 60-second exam cheat sheet

### Quantifiers

$\forall$  = every;  $\exists$  = at least one.

$\neg\forall P = \exists\neg P$ ;  $\neg\exists P = \forall\neg P$ .

### Compound negation

$\neg(P \wedge Q) = \neg P \vee \neg Q$ ;  $\neg(P \vee Q) = \neg P \wedge \neg Q$ ;  $\neg(P \rightarrow Q) = P \wedge \neg Q$ .

### Proof choice

Direct: assume  $p$ , derive  $q$ .

Contradiction: assume  $p$  and  $\neg q$ , derive impossible result.

Contrapositive: prove  $\neg q \rightarrow \neg p$ .

Counterexample: one value kills a universal claim.

Witness: one value proves an existential claim.

Vacuous: antecedent cannot occur.

### Do not mix up

Converse  $q \rightarrow p$  is not generally equivalent to  $p \rightarrow q$ . Contrapositive  $\neg q \rightarrow \neg p$  is equivalent.